

Bridge2CS User Manual

Printable guide for students, staff, and administrators using the Bridge2CS Student Portal

Website: <https://bridge2cs.onrender.com/#application-form>

Purpose: This manual explains how to use the Bridge2CS website step by step, including student applications, save draft/resume application, ZIP code eligibility, financial aid eligibility, email verification, CSV export planning, and MySQL data analysis.

Audience: High school applicants, program staff, admissions/recruitment staff, and database administrators.

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1. Website Overview

Bridge2CS is a centralized recruitment and application portal for Forsyth County and Winston-Salem students interested in computer science opportunities at Winston-Salem State University.

The website highlights academic readiness, mentorship, STEM exploration, scholarship/pathway support, student community, and application submission.

Main website features include:

- Home page with program information and Bridge2CS benefits
- Dark Mode toggle for a different visual display
- Start Application shortcut
- Save Draft shortcut
- Resume Saved Application option
- ZIP Code Eligibility Filter
- Student Application Form
- Financial Eligibility Check
- Email Verification tools
- Program goals and what-happens-next section

2. System Requirements

To use the website, users should have:

- A desktop, laptop, tablet, or smartphone
- A stable internet connection
- A modern web browser such as Chrome, Microsoft Edge, Firefox, or Safari
- JavaScript enabled in the browser
- Access to a valid email account for email verification

3. How to Use the Website

Follow these steps to access and move through the Bridge2CS website.

Step 1: Open a web browser.

Step 2: Go to <https://bridge2cs.onrender.com/#application-form>.

Step 3: Review the homepage information about the Bridge2CS program.

Step 4: Use the navigation buttons such as Start Application, Save Draft, or Resume Saved Application.

Step 5: Check ZIP code eligibility before completing the form.

Step 6: Complete the student application form.

Step 7: Use Save Draft if the application is not ready to submit.

Step 8: Complete the Financial Eligibility Check if financial support may be needed.

Step 9: Verify email if required.

Step 10: Submit the application and watch email for updates.

4. ZIP Code Eligibility Filter

The Eligibility Filter checks whether the applicant is from an eligible Forsyth County or Winston-Salem ZIP code before the student finishes the application.

How to use it:

1. Find the Eligibility Filter section.
2. Enter the applicant ZIP code.
3. Click **Check Eligibility**.
4. Read the message shown by the system.
5. Continue to the application only if the applicant is eligible.

Why this matters: The filter helps reduce incomplete or ineligible applications and keeps the program focused on its intended service area.

5. Student Application Form

The Student Application Form collects applicant information needed for recruitment, outreach, and program review.

Step-by-step application process:

1. Click **Start Application** or scroll to the Student Application Form.
2. Enter the student's full name.
3. Enter a valid email address.
4. Enter a phone number and SMS opt-in information if requested.
5. Enter the student's high school.
6. Enter the ZIP code.
7. Select the expected graduation year.
8. Answer whether the student is first-generation.
9. Select interest indicators such as Computer Science, IT, Data Science, Engineering, Mathematics, Robotics, or AI/ML.
10. Type a short response explaining what motivates the student to join.
11. Review all information for accuracy.
12. Click **Submit Application** to send the application.

Tip: Students should use a working email address because program updates may be sent by email.

6. Save Draft and Resume Saved Application

The website includes a Save Draft feature that allows applicants to pause and continue later. Saved drafts are intended for short-term continuation, about 30 days.

To save an application draft:

1. Begin filling out the Student Application Form.
2. Enter as much information as possible.
3. Click **Save Draft** instead of Submit Application.
4. Confirm that the draft has been saved.
5. Return later to continue.

To resume a saved application:

1. Return to the Bridge2CS website.
2. Click **Resume Saved Application**.
3. Enter the required identifying information if prompted.

4. Review the restored application details.
5. Update missing or incorrect information.
6. Click **Submit Application** when finished.

Important: Save Draft does not replace submitting the application. The application is not complete until Submit Application is selected.

7. Financial Aid Eligibility Check

The Financial Eligibility Check is a FAFSA-based estimator that gives students a quick idea of possible program financial support eligibility. It is not an official financial aid determination.

How to use the Financial Eligibility Check:

1. Scroll to the Financial Eligibility Check section.
2. Select whether the FAFSA has been completed.
3. Select Pell Grant eligibility: Yes, No, or Unsure.
4. Select the estimated household income range.
5. Select first-generation college student status.
6. Click **Check Financial Eligibility**.
7. Read the result carefully.

What the estimator may use: FAFSA completion, Pell Grant eligibility, household income range, and first-generation status.

Reminder: Final financial aid decisions should be confirmed through official school or financial aid office processes.

8. Email Verification

The Email Verification section helps confirm that the applicant can receive messages from the program.

How to verify email:

1. Enter the applicant email address if prompted.
2. Click **Send Verification Code**.
3. Check the email inbox for the verification code.
4. Enter the code into the website.
5. Click **Confirm Verification**.
6. Continue with the application or submission process.

9. What Happens After Applying

After submitting the application, the website explains the general review process:

1. The student submits the application.
2. The Bridge2CS team reviews the information.
3. The student checks email for updates, guidance, or next steps.

Applicants should check spam/junk folders if they do not see emails from the program.

10. CSV Export Fields

Applicant data should be exportable into a CSV file so staff can import it into MySQL and generate useful statistics.

The CSV should include these columns:

CSV Column	Meaning
full_name	Applicant's full name
email	Applicant's email address
phone_number	Applicant's phone number
high_school	Applicant's current high school
zip_code	Applicant ZIP code
expected_graduation_year	Expected high school graduation year
first_gen	First-generation college student status
major_interest	Primary academic or STEM interest

Example CSV layout:

```
full_name,email,phone_number,high_school,zip_code,expected_graduation_year,first_gen,major_interest
John Doe,johndoe@email.com,3365551234,Parkland High School,27103,2027,Yes,Computer Science
Jane Smith,janesmith@email.com,3365555678,Reagan High School,27106,2026,No,Cybersecurity
Michael Brown,michael@email.com,3365559999,East Forsyth High School,27101,2027,Yes,Data Science
```

11. MySQL Setup and CSV Import

This section guides staff through preparing MySQL to store and analyze Bridge2CS applicant data.

Step 1: Create the database

```
CREATE DATABASE bridge2cs;
USE bridge2cs;
```

Step 2: Create the applicants table

```
CREATE TABLE applicants (
  applicant_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL,
  phone_number VARCHAR(20),
  high_school VARCHAR(150),
  zip_code VARCHAR(10),
  expected_graduation_year INT,
  first_gen VARCHAR(10),
  major_interest VARCHAR(100)
);
```

Step 3: Save the CSV file

Save the exported CSV file as **applicants.csv**. Keep the first row as the header row.

Step 4: Import the CSV file into MySQL

```
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/applicants.csv'
INTO TABLE applicants
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
```

```
(full_name, email, phone_number, high_school, zip_code, expected_graduation_year, first_gen,  
major_interest);
```

Windows/XAMPP note: If LOAD DATA INFILE does not work, use phpMyAdmin: choose the database, select the applicants table, click Import, choose the CSV file, and match the columns to the table fields.

Step 5: Confirm the data imported

```
SELECT * FROM applicants LIMIT 10;
```

12. SQL Queries for Program Statistics

The following queries help administrators find specific applicant groups and program trends.

View all applicants

```
SELECT * FROM applicants;
```

Count total applicants

```
SELECT COUNT(*) AS total_applicants FROM applicants;
```

Fetch only first-generation applicants

```
SELECT * FROM applicants  
WHERE first_gen = 'Yes';
```

Count first-generation applicants

```
SELECT COUNT(*) AS first_gen_total  
FROM applicants  
WHERE first_gen = 'Yes';
```

Fetch applicants from a specific ZIP code

```
SELECT * FROM applicants  
WHERE zip_code = '27103';
```

Count applicants by ZIP code

```
SELECT zip_code, COUNT(*) AS total  
FROM applicants  
GROUP BY zip_code  
ORDER BY total DESC;
```

Fetch applicants by high school

```
SELECT * FROM applicants  
WHERE high_school = 'Parkland High School';
```

Count applicants by high school

```
SELECT high_school, COUNT(*) AS total  
FROM applicants  
GROUP BY high_school  
ORDER BY total DESC;
```

Fetch applicants by expected graduation year

```
SELECT * FROM applicants  
WHERE expected_graduation_year = 2027;
```

Count applicants by graduation year

```
SELECT expected_graduation_year, COUNT(*) AS total  
FROM applicants  
GROUP BY expected_graduation_year  
ORDER BY expected_graduation_year;
```

Fetch applicants interested in Computer Science

```
SELECT * FROM applicants  
WHERE major_interest = 'Computer Science';
```

Count applicants by major interest

```
SELECT major_interest, COUNT(*) AS total
FROM applicants
GROUP BY major_interest
ORDER BY total DESC;
```

Find duplicate emails

```
SELECT email, COUNT(*) AS total
FROM applicants
GROUP BY email
HAVING COUNT(*) > 1;
```

Search by part of a student name

```
SELECT * FROM applicants
WHERE full_name LIKE '%Smith%';
```

Search by phone number

```
SELECT * FROM applicants
WHERE phone_number = '3365551234';
```

Optional Financial Aid Data Table

If the website stores financial eligibility answers separately, staff can create a related table using applicant_id as the connection to the applicants table.

```
CREATE TABLE financial_aid_checks (
  check_id INT AUTO_INCREMENT PRIMARY KEY,
  applicant_id INT,
  fafsa_completed VARCHAR(10),
  pell_eligible VARCHAR(10),
  household_income_range VARCHAR(50),
  first_gen VARCHAR(10),
  estimated_result VARCHAR(50),
  FOREIGN KEY (applicant_id) REFERENCES applicants(applicant_id)
);
```

Example financial aid queries:

```
SELECT * FROM financial_aid_checks
WHERE pell_eligible = 'Yes';

SELECT household_income_range, COUNT(*) AS total
FROM financial_aid_checks
GROUP BY household_income_range
ORDER BY total DESC;

SELECT a.full_name, a.email, f.estimated_result
FROM applicants a
JOIN financial_aid_checks f ON a.applicant_id = f.applicant_id
WHERE f.estimated_result = 'Potentially Eligible';
```

13. Troubleshooting

Issue	Possible Fix
Website will not load	Check internet connection, refresh the page, or try another browser.
Eligibility check does not work	Make sure the ZIP code is entered correctly.
Application will not submit	Check that required fields are completed and email is valid.
Draft is missing	Saved drafts are short-term; check that the same device/browser is being used if local storage is used.
Verification code not received	Check spam/junk folder and confirm the email address is correct.
CSV import fails	Check column order, header row, commas, line endings, and MySQL file permissions.
Phone numbers lose leading zeros or formatting	Store phone numbers as VARCHAR, not INT.
ZIP codes lose leading zeros	Store ZIP codes as VARCHAR, not INT.

14. Privacy and Security

Bridge2CS stores student application information, so staff should handle all data carefully.

- Only authorized staff should access exported CSV files.
- Do not share applicant CSV files through unsecured channels.
- Keep financial aid information private.
- Use strong database passwords.
- Back up the database regularly.
- Remove duplicate or test records before generating official statistics.
- Do not publish student emails, phone numbers, or financial details.

Final Notes

This manual should be updated whenever the website adds new features, changes application fields, or modifies the CSV export format. Staff should test the full process from application submission to CSV export and MySQL analysis before using the system for official reporting.

15. Admin Dashboard Overview

The Admin Dashboard is used to manage recruitment data, applicant tracking, analytics, communication tools, and reporting for the Bridge2CS platform.

Feature	Description
Admin Login System	Secure administrator authentication portal
Dashboard Metrics & Analytics	Displays recruitment and engagement analytics
CSV Upload & R Analytics	Uploads applicant datasets and runs analytics
Communication Tools	Sends campaign emails and reminders
Visual Analytics Charts	Displays analytics charts and reports
Enrollment Funnel	Tracks applicant progression stages
Geographic Visualization	Displays applicant geographic distribution
Data Export Features	Exports CSV and JSON summaries

16. Admin Login & Dashboard Usage

Administrators access the dashboard through the secure login page.

Login Steps:

1. Open the Admin Dashboard.
2. Enter administrator username.
3. Enter administrator password.
4. Click **Login**.

Default Credentials
Username: admin
Password: bridge123

Step-by-Step Dashboard Usage:

1. Review dashboard metrics and applicant analytics.
2. Upload applicant CSV datasets.
3. Run R Analytics for statistical analysis.
4. Download filtered CSV reports.
5. Download JSON summaries.
6. Use communication tools to send campaign emails and reminders.
7. Monitor applicant progression through the enrollment funnel.
8. Review geographic visualizations and interest area charts.

17. Dashboard Features & Technical Overview

Enrollment Funnel Stages:

- Submitted
- Under Review
- Accepted
- Enrolled

Visual Analytics Includes:

- Application stages

- Engagement metrics
- Geographic distribution
- Interest area breakdown

Component	Technology
Frontend	HTML / CSS / JavaScript
Backend	Node.js / Express
Database	SQLite
Hosting	Render

Key Files:

- server.js
- script.js
- admin.html

R Analytics Deployment Explanation:

The R analytics pipeline currently runs locally because the Render free-tier hosting environment does not support R runtime environments or Docker containers.